

Total No. of Questions : 6]

SEAT No. :

P87

BE/Insem./APR-127

[Total No. of Pages : 1

B.E. (Mechanical Engineering)

**402050 C : DESIGN OF PUMPS, BLOWERS AND COMPRESSORS
(2012 Pattern) (Semester - II) (Elective - IV)**

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6.
- 2) Figures to the right indicate full marks.
- 3) Use of calculator is allowed.
- 4) Assume suitable data, if necessary.
- 5) Neat diagrams must be drawn wherever necessary.

Q1) a) What is specific speed? Explain its significance. [5]

b) Explain performance characteristics of a reciprocating compressor. [5]

OR

Q2) a) Write short note on stage velocity triangles. [5]

b) Differentiate between compressible and Incompressible flow machines. [5]

Q3) a) Explain construction of double acting reciprocating pump with neat sketch. [5]

b) Explain the effect of acceleration in suction and delivery pipes on indicator diagram. [5]

OR

Q4) a) What are different applications of reciprocating pumps? Explain any one in detail. [5]

b) Explain air vessel in reciprocating pump. [5]

Q5) a) Explain design procedure of a centrifugal pump. [5]

b) Explain the following terms. [5]

- i) Static suction head
- ii) Static Discharge head.
- iii) Monometric head.

OR

Q6) a) Explain various types of pump casing. [5]

b) Write short note on pump selection. [5]

